Production, Productivity & Efficiency

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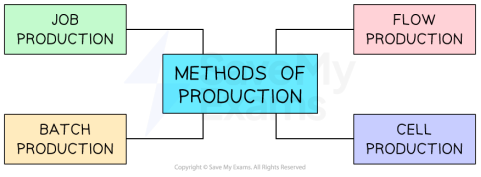
Methods of production

- Production is the **transformation of resources** (e.g. raw materials, components and processes) into finished goods or services
  - Goods are **physical products**, such as bicycles and T-shirts
  - Services are **non-physical items**, such as hairdressing, tourism and manicures
- Businesses can **organise their production** processes in a variety of ways

The main methods of production



The main methods of production are job, batch, flow and cell production

- The method of production used by a business will depend on a number of factors, including:
  - The **level of output** required to be produced
  - The **nature** of the product
  - Whether the product is **standardised or customised**
  - The level of **automation** used in production

Job production

- Producing **one item at a time**, as ordered by the customer
  - Advantages
    - **High-quality** product
    - Motivated and **highly skilled workers**
    - Customised products can be produced
  - Disadvantages
    - Production is **slow**
    - **Labour costs** are high

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Batch production

        - Workers can **specialise**
        - Production can take place as the previous batch starts running out
    - Disadvantages
        - **Requires careful coordination** to avoid shortages
        - Money is **tied up in stock**, as completed products need to be stored

Flow production

- **Continuous manufacturing** of standardised products, usually on a production line
    - Advantages
        - Low unit costs due to **economies of scale**
        - Rapid production
        - Usually highly automated (**capital-intensive**)
    - Disadvantages
        - Customisation is **difficult**
        - Capital equipment can be **expensive to purchase**

Cell production

- This involves workers being organised into **multi-skilled teams**, with each team responsible for a **particular part of the production process**
    - Advantages
        - Cell production is often more efficient than other methods, as workers share their skills and expertise
        - Motivation is usually high, as employees work as a team
    - Disadvantages
        - Requires extensive reorganisation of production processes
        - Teams' efficiency may be reduced by weaker workers

Worked Example

Blush Cosmetics uses batch production to make its range of soaps and bath bombs, with a variety of unique ingredients and scents. The business sells its products in its own high street store and online and has recently started supplying its products in small quantities to a chain of exclusive hotels.

Explain one likely reason why Blush Cosmetics chooses to use batch production. [4]

Step 1: Identify a reason for the business to use batch production

One reason for Blush Cosmetics to use batch production is that it allows groups of products with varied ingredients to be

Blush Cosmetics sells a range of bath bombs and soaps that have different fragrances, colours and ingredients. **[Ap]**

Step 3: Develop the reason using a connective

Blush Cosmetics needs to produce significant levels of output to meet increasing demand and batch production, as well as to provide the opportunity to change ingredients between batches **[An]**, which can allow quite large quantities to be produced for the company's high street and online stores **[An]**.

Or:

Using batch production can allow Blush Cosmetics to produce smaller quantities of unique products for exclusive hotel customers **[Ap]**, meeting their specific needs **[An]**.

Examiner Tips and Tricks

Carefully consider the needs of the customers to whom a business sells when recommending a suitable method of production.

Where the selling price is a key driver of consumer demand, flow production (where **unit costs** are minimised) is likely to be very suitable.

Where demand is driven by quality or where customisation is required, job production or batch production is likely to be a better choice.

Calculating productivity

- **Productivity** is the output per input (person or machine) per hour (e.g. an IKEA worker is able to produce two Poäng chairs per hour)
  - This is not production, which is the **total amount of output produced** in a time period (e.g. IKEA produced 300 Poäng chairs in February)

Labour productivity

- The **labour productivity** of a business is a measure of the **output per worker** during a specified period of time
- Labour productivity is calculated using the formula

$$\text{Labour productivity} = \frac{\text{Output}}{\text{Number of workers}}$$

Capital productivity

- **Capital productivity** is a measure of the **output of capital employed (e.g. machinery)** during a specified period of time
- Capital productivity is calculated using the formula

Worked Example

The table shows the number of pairs of luxury wool socks produced by Scotty Socks Ltd in 2021 and 2022.

Year	Units produced
2021	46,000
2022	69,000

In 2021, Scotty Socks employed 50 staff. In 2022, the number of staff employed by the business increased by 20%.

Calculate the percentage change in labour productivity between 2021 and 2022.

[4]

Step 1: Calculate the labour productivity for 2021

$$= \frac{46{,}000 \text{ units}}{50 \text{ workers}} \quad [1]$$

$$= 920 \text{ units per worker}$$

Step 2: Calculate the labour productivity for 2022

$$= \frac{69{,}000 \text{ units}}{60 \text{ workers}} \quad [1]$$

$$= 1{,}150 \text{ units per worker}$$

Step 3: Calculate the percentage difference between the two years ([new−old] ÷ old)

$$= \frac{1{,}150 - 920 \text{ units}}{920 \text{ units}} \times 100 \quad [1]$$

$$= 25\%$$

Step 4: Identify whether the percentage difference is an increase or a decrease

25% increase [1]

Worked Example

Rolvo Ltd forecasts that by the end of the year, it will produce 250,800 units and that capital productivity will be 1,100 units per machine.

Calculate the number of machines Rolvo Ltd has in use.

Step 1: Divide the output by the capital productivity

Factors that influence productivity

- The productivity within a business can often be improved
- When productivity increases, **business costs decrease**
- When business costs decrease, the firm can either pass on this decrease to consumers in the form of **lower prices** or maintain the selling prices and enjoy higher **profit margins**

Factors that influence productivity

Factor	Explanation

Employee motivation	• Motivated workers tend to be more productive
	  ○ **Financial incentives linked to output** may increase worker productivity
	  ○ A non-financial incentive may be **including workers in decision-making**, which will increase their commitment and productivity

Skills, education and training staff	• Well-trained and educated workers are likely to be **able to make useful contributions** to decisions that improve productivity
	  ○ Workers are more **autonomous**, so the need for supervision is reduced
	  ○ Contributions from knowledgeable staff are likely to lead to **improvements in productivity**

Business organisation and working practices	• Flexible and adaptable workplaces can **improve the commitment of workers** and allow a business to **respond to changes in demand** • **Hours and locations of work** can be adapted to better meet the needs of workers and the demand • **Workstations may be used for various purposes** with careful planning and training
Investment in capital equipment	• Increased automation can **improve levels of output and quality** ○ Well-chosen machinery is **less likely to make mistakes** than humans

	are properly maintained

The link between productivity and competitiveness

- Competitiveness refers to the ability of a business to maintain or grow its sales and market share, **given the presence and actions of rivals**
- Businesses that increase their level of productivity (e.g. of workers or capital equipment) are likely to be **more competitive**

Productivity and competitiveness

There is a strong link between productivity and business competitiveness

- Businesses that are competitive are likely to have the **financial resources** required to continue investing in improvements to their productivity

Understanding efficiency

- Efficiency refers to the ability of a business to **use its production resources** as cost-effectively as possible
  - Efficiency is often measured in terms of the **average cost per unit**
  - The average cost per unit is calculated using the formula

$$\text{Average cost per unit} = \frac{\text{Total costs}}{\text{Number of units}}$$

- **Maximum efficiency** is achieved when the cost per unit is at its lowest

Diagram showing maximum efficiency

Graph showing that maximum efficiency occurs at the point where the costs are at their lowest

- The average cost curve shows that the most efficient level of production is achieved when:
  - **Economies of scale** are maximised
  - Total costs are spread across an optimum level of output
  - **Diseconomies of scale** are minimised

Factors that influence efficiency

Factors that can influence business efficiency

Factor	Explanation
Standardisation of the production process	• Occurs when all staff use the **same components and techniques** in the production process ○ **Training of workers** is minimised ○ **Bulk buying** of components reduces variable costs ○ Production **lead time is reduced** ○ **But** customisation of products is not usually possible
Relocation or downsizing	• Moving production to a **cheaper or smaller location** can reduce **fixed costs** ○ **Labour-intensive** businesses may look for **lower-wage locations** ○ Having capital-intensive locations may mean looking for **lower rents or land costs** ○ However, relocation is very disruptive and will incur significant short-term costs

	increase the rate of output, lower costs and improve quality

Organisational restructuring	• **Reducing the level of staff** or **reorganising** staff can better match labour to output needs
	  ○ **Delayering reduces labour costs**, as levels of management are removed
	  ○ Redeployment can **motivate workers** by providing opportunities for staff to take on a new role, which will develop their skills and experience

Outsourcing	• Tasks may be given to **other specialist businesses** that can complete them at a lower cost ○ Outsourcing allows a business to focus on improving the efficiency of its **core competencies**

Adoption of lean production techniques	• An approach to production that involves the **reduction of all types of wastage** (time, resources and space)
	  ○ **Kaizen** means that **improvements are made continuously**
	  ○ **Just-in-time** involves **holding little or no stock**, which minimises storage costs

Capital-intensive and labour-intensive production

- **Labour-intensive production** predominantly **uses physical labour** in the production of goods/services
  - The **delivery of services** is usually more labour-intensive than manufacturing
  - In countries where **labour costs are low** (e.g. Bangladesh), labour-intensive production is common
  - **Small-scale production** is likely to be labour-intensive
  - For example, UK schools are labour-intensive operations, as teachers plan and deliver lessons and provide pastoral support
- **Capital-intensive production** predominantly **uses machinery and technology** in the production of

- Manufacturing in **developed countries**, where **labour costs are relatively high**, is likely to be capital-intensive
- For example, automotive manufacturers such as Ford use robots and other production technology to manufacture vehicles, with supervisors overseeing the quality of output

Evaluation of labour-intensive and capital-intensive production

Type of production	Advantages	Disadvantages
Capital-intensive	• **Low-cost production** where output is high • Machines are usually consistent and precise • Machines can **run without breaks**	• Significant set-up and **maintenance costs** • **Breakdowns** can severely delay production • May not provide **flexibility** in production
Labour-intensive	• **Low-cost production** where labour costs are low • Provides opportunities for workers to be creative • Workers are flexible (e.g. they can be retrained)	• Workers **may be unreliable** and need regular breaks • **Incentives** may be needed to motivate staff • **Training costs** can be significant

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